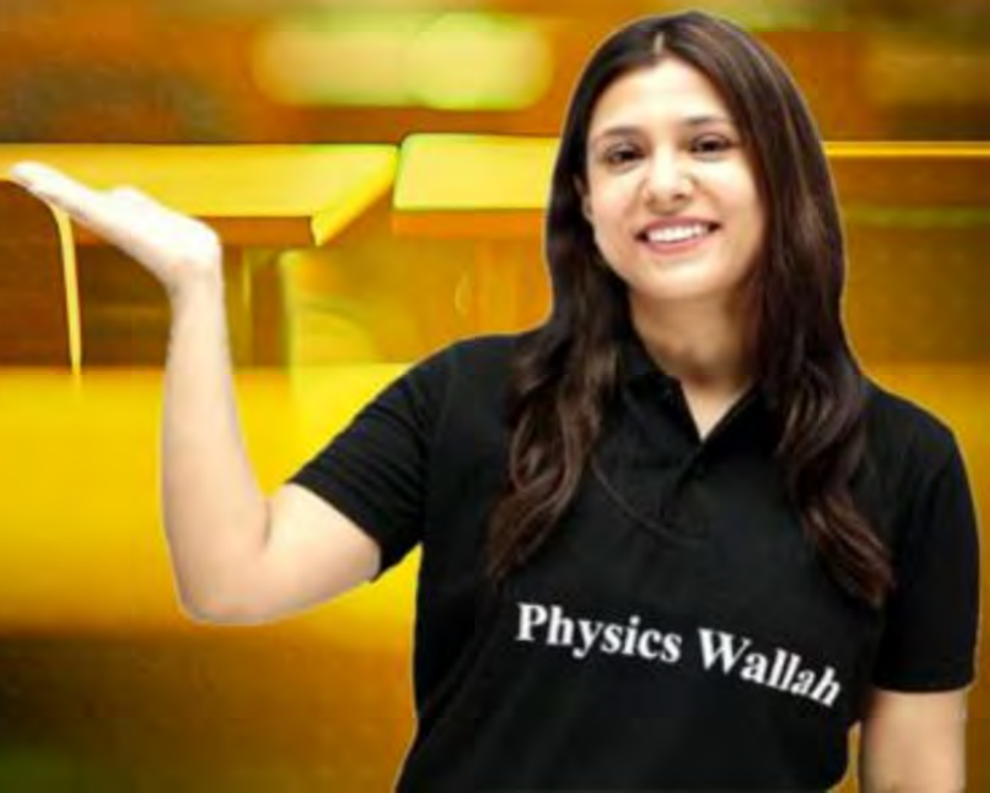
RADIANT 2026

Biology

Respiration in Plants

Lecture - 01

By - Upasana Mongia Ma'am



Topics *to be covered*

1. Respiration and its Types
2. Aerobic Respiration
3. [redacted]



What is Respiration?

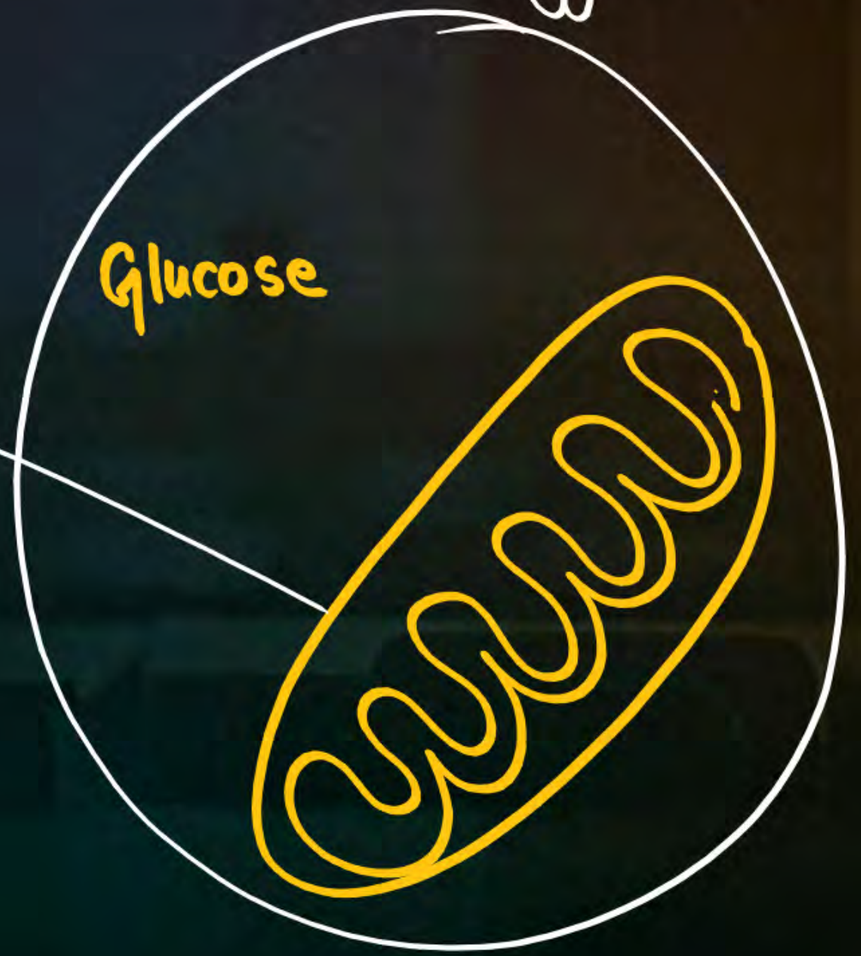
ANIMALS

Carbohydrates → Digestion → Glucose

$C_6H_{12}O_6$



Duggu's Cell



What is Respiration

- All living cells in a plant, as well as those in animals, require energy for various activities of the body.



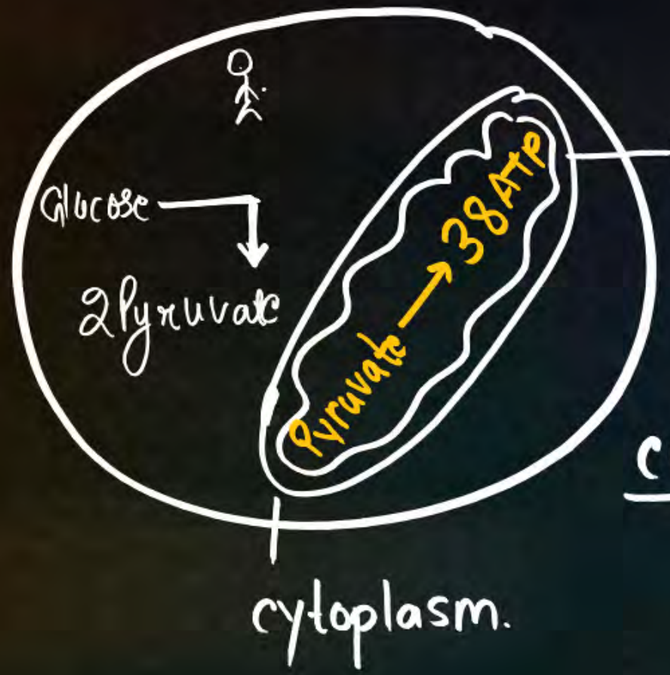
G+G+G+G.....

- For Example, building up proteins from amino acids, making starch from glucose. Absorbing minerals from the soil, or laying down cell walls by plant cells, are all such activities that require energy.

- This energy is made available by the breakdown of glucose, a simple carbohydrate.

$$C_6H_{12}O_6 + 6O2 \xrightarrow{\text{Enzyme}} 6CO_2 + 6H_2O + \text{Energy}$$

(ATP)



Glyco → Glucose

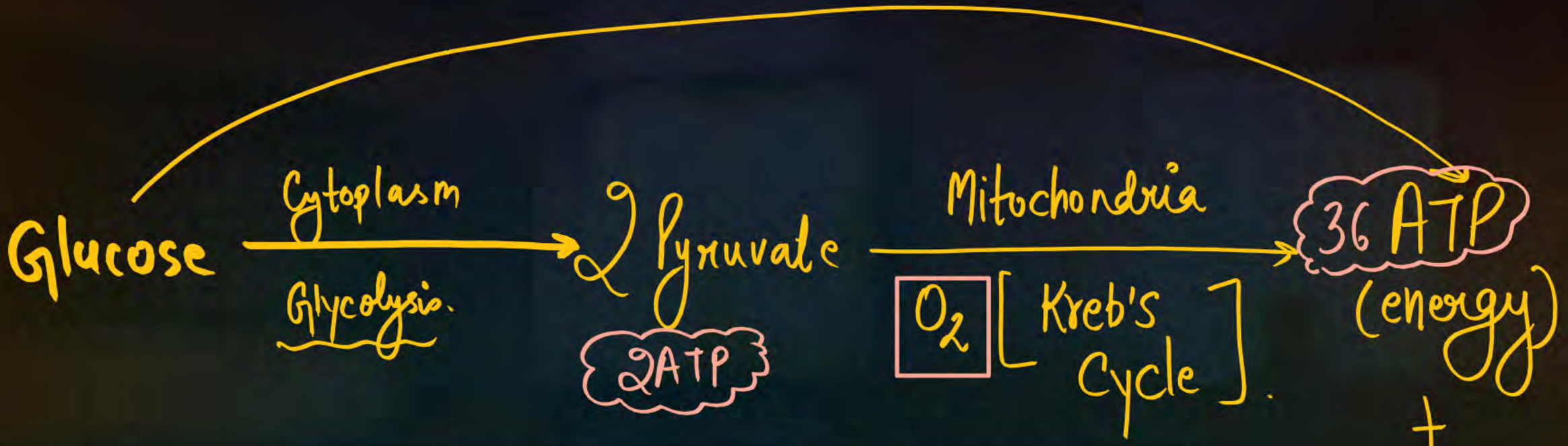
Lysis → to Break

$$\text{Glucose} \xrightarrow[\text{GLYCOLYSIS}]{\text{Cytoplasm}} \underline{\text{2 Pyruvate}} \xrightarrow[O_2\ [\text{Krebs Cycle}]]{\text{Mitochondria}} \text{ATP (Energy)} + CO_2$$

or

Pyruvic acid

$$C_6H_{12}O_6 \longrightarrow 2C_3H_4O_3 \longrightarrow$$



AEROBIC RESPIRATION

- When the glucose is broken down in the presence of O_2 to give [38 molecules of ATP] along with CO_2 and H_2O is called Aerobic Respiration.

ATP

- Adenosine Triphosphate (ATP)
- Adenosine Diphosphate (ADP)
- Adenosine Monophosphate (AMP)
  - It is called Energy currency of the cell, Energy is released when the bonds are broken, and ATP is converted to ADP and further into AMP.

ATP

Conversion of Glucose into Pyruvate.—

A) Glycolysis

B) Respiration

C) Citric acid cycle

D) Kreb's Cycle

METABOLISM

Living organisms show types of metabolic activities :

(i) **Anabolic** (constructive or biosynthetic processes), it consumes energy. When molecules combine together to form large molecule. Eg Glycogen formation from Glucose.

(ii) **Catabolic** (destructive or breaking down processes), it gives out energy for use by the organism. e.g Respiration

- Respiration is a breaking down process by which a living cell oxidises organic substances (glucose) and releases carbon dioxide, water and energy.

in animals G+G+G+G+G ⟶ (Glycogen) liver

in plants G+G+G+G ⟶ Starch

Catabolic process is also known as Biosynthetic process.

A) TRUE

B) FALSE.

Respiration is a Catabolic Process.

A) TRUE

B) FALSE.

The stored food material in plants is ___

A) Glucose

B) Glycogen

C) Amino acids

D) Starch.

Respiration Vs Burning (Combustion)

Respiration	Burning/Combustion
1. Cellular process	Non-cellular process
2. Occurs at body temperature	Occurs at high temperature (at ignition point)
3. Occurs in a series of chemical steps	Occurs in a single step
4. Carried out by enzymes	Carried out by heat
5. Biochemical process	Physico-chemical process
6. Energy is liberated as ATP and some heat	All the energy is liberated as heat and light.
7. No light energy is produced	Light energy is produced

The Entire Plant Respires

- Every part of a plant such as the leaves, stem, roots, and even the deepest cell placed in any region respires. Oxygen in plants is obtained from the atmosphere through three inlets:
- **Stomata in leaves**
- **Lenticels in stem**
- **General surface of the roots**

Major differences between aerobic and anaerobic respiration in plants

Aerobic respiration	Anaerobic respiration
1. Proceeds in the presence of oxygen.	Proceeds without using oxygen.
2. Complete breakdown of glucose.	Incomplete breakdown of glucose.
3. End-products are carbon dioxide and water.	End-products are ethyl alcohol and carbon dioxide.
4. Energy liberated in large quantity (38 ATP) from one mole of glucose.	Energy liberated in small quantity (2 ATP) from one mole of glucose.
5. Occurs normally throughout life.	Occurs temporarily for short periods.


Thank You